WORKED SOLUTION

STEP II 2020, Question 10

Mechanics • Energy and Equilibrium

STEP II 2020, Question 10

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

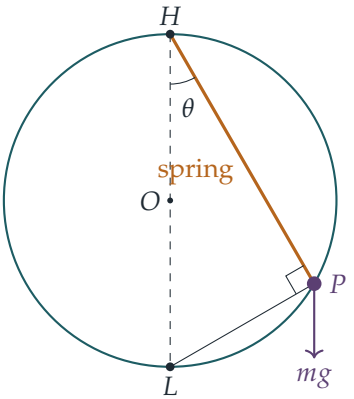
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The key idea

Because HL is a diameter, the angle in a semicircle gives the spring length and the height of P directly in terms of θ: $HP = 2a\cos\theta$, and P is $2a\cos^2\theta$ below H. The wire is smooth, so its reaction does no work. That makes the whole problem about one function, the potential energy as a function of $c = \cos\theta$. It turns out to be a quadratic in c, and completing the square gives both parts at once.

Solution

Model P as a particle on a smooth wire, and the spring as light and obeying Hooke's law (in tension or compression). Let O be the centre of the circle.



Geometry. Since HL is a diameter, $\angle HPL = 90°$, so in the right-angled triangle HPL with hypotenuse $2a$,

$$HP = 2a\cos\theta.$$

The vertical drop from H to P is $HP\cos\theta = 2a\cos^2\theta$. As P goes round from L towards H on either side, θ runs over $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\cos\theta > 0$.

Potential energy. Take H as the level of zero gravitational potential energy and write $c = \cos\theta$. The extension of the spring is $2ac - l$, so

$$\begin{aligned} V &= -2mga\,c^2 + \frac{\lambda}{2l}(2ac - l)^2 \\ &= \frac{2a(a\lambda - mgl)}{l}\,c^2 - 2a\lambda\,c + \frac{\lambda l}{2}. \end{aligned}$$

Part (i)

The reaction of the smooth wire does no work, so P is in equilibrium exactly where $\mathrm{d}V/\mathrm{d}\theta = 0$. (Equivalently, the forces along the tangent balance.) By the chain rule,

$$\frac{\mathrm{d}V}{\mathrm{d}\theta} = -\sin\theta\,\frac{\mathrm{d}V}{\mathrm{d}c} = -\sin\theta\left(\frac{4a(a\lambda - mgl)}{l}\,c - 2a\lambda\right).$$

At $\theta = \alpha$ with $0 < \alpha < \frac{\pi}{2}$ we have $\sin\alpha \neq 0$, so the bracket must vanish:

$$\frac{4a(a\lambda - mgl)}{l}\cos\alpha = 2a\lambda.$$

The right-hand side is non-zero, so $a\lambda \neq mgl$, and we can divide:

$$\cos\alpha = \frac{\lambda l}{2(a\lambda - mgl)},$$

as required.

For such an α to exist we need $0 < \cos\alpha < 1$.

- $\cos\alpha > 0$: the numerator λl is positive, so we need $a\lambda > mgl$.
- $\cos\alpha < 1$: multiplying up by the positive quantity $2(a\lambda - mgl)$ gives $\lambda l < 2a\lambda - 2mgl$, that is, $\lambda(2a - l) > 2mgl$.

Since $l < 2a$, the second condition is $\lambda > \dfrac{2mgl}{2a - l}$. This also implies the first, because $\dfrac{2mgl}{2a - l} > \dfrac{2mgl}{2a} = \dfrac{mgl}{a}$. So an equilibrium with $\alpha > 0$ exists only if $\lambda > \dfrac{2mgl}{2a - l}$. (Conversely, if this holds, the formula gives a genuine angle $\alpha \in (0, \frac{\pi}{2})$, and by symmetry $-\alpha$ is an equilibrium too.)

Part (ii)

> **?** **Ask yourself** Is there a neat way to write V that makes $\cos\alpha$ appear by itself?

Yes: complete the square. Write $K = \dfrac{2a(a\lambda - mgl)}{l}$. From part (i), $\dfrac{a\lambda - mgl}{l} = \dfrac{\lambda}{2\cos\alpha}$, so

$$K = \frac{a\lambda}{\cos\alpha}, \qquad \frac{2a\lambda}{K} = 2\cos\alpha.$$

Hence

$$V = K\left(c^2 - 2c\cos\alpha\right) + \frac{\lambda l}{2} = \frac{a\lambda}{\cos\alpha}(c - \cos\alpha)^2 + \text{constant}.$$

(As a check, this has its minimum at $c = \cos\alpha$, the equilibrium found in part (i).) We assume, as part (i) requires for the equilibrium to exist, that $a\lambda > mgl$, so $\cos\alpha > 0$ and $K > 0$.

At L, $\theta = 0$ and $c = 1$. Conservation of energy between L and a general point gives

$$\tfrac{1}{2}mv^2 = \tfrac{1}{2}mu^2 + \frac{a\lambda}{\cos\alpha}\Big[(1 - \cos\alpha)^2 - (c - \cos\alpha)^2\Big].$$

Multiplying by $\dfrac{\cos\alpha}{a\lambda} > 0$,

$$\frac{mv^2\cos\alpha}{2a\lambda} = D - (c - \cos\alpha)^2, \qquad \text{where } D = (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha. \qquad (*)$$

So if P comes to rest at $\theta = \beta$, then $v = 0$ there and

$$(\cos\alpha - \cos\beta)^2 = (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha,$$

which is the required equation.

Ask yourself The equation has two roots for $\cos\beta$. Which one is the actual stopping point, and when is it before H?

As P rises from L towards H, c decreases from 1 towards 0. By $(*)$, P keeps moving as long as $(c - \cos\alpha)^2 < D$, so it first stops where c meets the nearer end of the interval $(\cos\alpha - \sqrt{D},\ \cos\alpha + \sqrt{D})$ when moving down from 1. Now $D \geq (1 - \cos\alpha)^2$ (strictly if $u > 0$), so $\cos\alpha + \sqrt{D} > 1$ and $\cos\alpha - \sqrt{D} < 1$. Thus $c = 1$ lies inside the interval, and the stopping value is

$$\cos\beta = \cos\alpha - \sqrt{D}.$$

P stops before reaching H (where $c = 0$) exactly when this is positive, that is, when $\sqrt{D} < \cos\alpha$, or $D < \cos^2\alpha$:

$$\begin{aligned} & (1 - \cos\alpha)^2 + \frac{mu^2}{2a\lambda}\cos\alpha < \cos^2\alpha \\ \iff \quad & 1 - 2\cos\alpha + \frac{mu^2}{2a\lambda}\cos\alpha < 0 \\ \iff \quad & \frac{mu^2}{2a\lambda} < 2 - \sec\alpha, \end{aligned}$$

dividing by $\cos\alpha > 0$ in the last step. Hence P comes to rest before H only if

$$u^2 < \frac{2a\lambda}{m}(2 - \sec\alpha).$$

In particular this needs $\sec\alpha < 2$, i.e. $\alpha < \frac{\pi}{3}$. Otherwise P reaches H whatever its speed at L.

Checks

- Take $m = 1$, $g = 10$, $a = 1$, $l = 1$, $\lambda = 40$ (in consistent units). Then $\lambda > 2mgl/(2a - l) = 20$ and $\cos\alpha = 40/60 = \frac{2}{3}$. The critical value is $u^2 = 80(2 - \frac{3}{2}) = 40$. With $u^2 = 40$, $D = \frac{1}{9} + \frac{40}{80} \cdot \frac{2}{3} = \frac{4}{9}$, so $\cos\beta = \frac{2}{3} - \frac{2}{3} = 0$. The particle just reaches H, as it should.
- If $u = 0$ then $D = (1 - \cos\alpha)^2$ and $\cos\beta = 1$. A particle at rest at L stays there, which is right because L is also an equilibrium.
- A computer integration of the equation of motion, for many random parameter sets, stopped the particle at the predicted $\cos\beta$ exactly when u^2 was below the critical value.

Common slips

- Taking $HP = 2a\sin\theta$, or measuring the height from O with the wrong angle. Remember that the angle at the centre is 2θ, not θ.
- Only checking $\cos\alpha > 0$ in part (i). It is the condition $\cos\alpha < 1$ that gives the bound on λ.
- Choosing the root $\cos\alpha + \sqrt{D}$ for $\cos\beta$. It exceeds 1, so it isn't a position on the wire.

Beyond STEP

Writing the energy as $K(c - \cos\alpha)^2 + \text{constant}$ is the one-dimensional version of a key idea in undergraduate mechanics. For a system with one degree of freedom, the equilibria are the stationary points of the potential, their stability depends on whether those points are minima or maxima, and the regions of allowed motion are the places where the energy exceeds the potential. Here, when $\alpha > 0$ exists, L becomes a maximum of V and so an unstable equilibrium. The spring makes the bottom of the wire unstable! Kleppner and Kolenkow, *An Introduction to Mechanics*, develops energy diagrams and stability in this spirit.

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